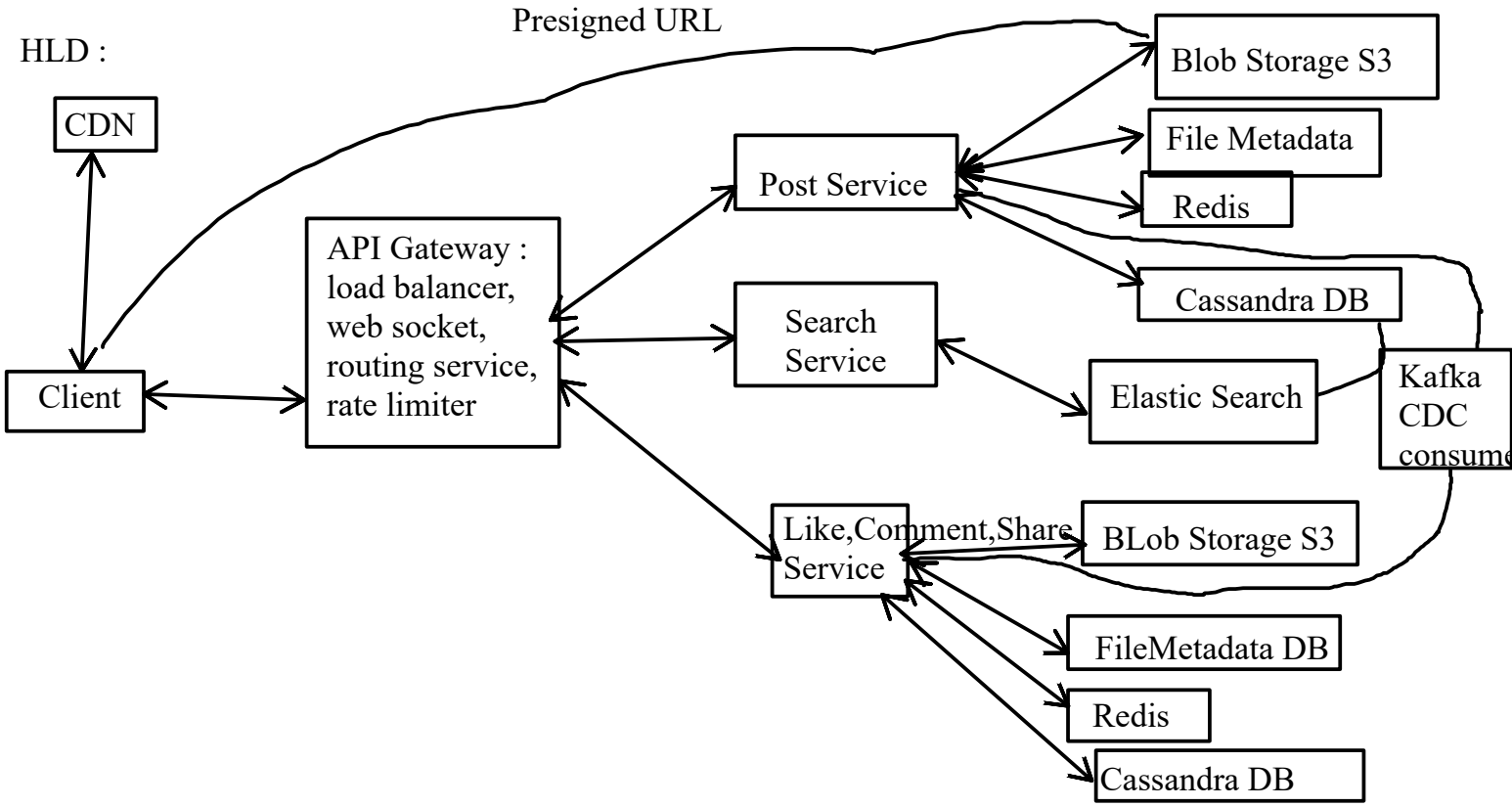
Facebook

Functional requirements:
user should be able to post images based on his preference,
user should able to access feed with updated posts,
user should be able to search for some post among the existing posts

Non-Functional requirements:
Data-partitionin,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem(here eventual consistency is fine , but here we need 99.99% uptime i.e AVailability>>Consistency)

Core Entities:
user,
post,
filemetadata,
history,

API :
POST:/upload/posts
{
userId,
{FileMetadata}
}
GET:search/{postID}

HLD :